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2.3.6 Best rank-k approximation

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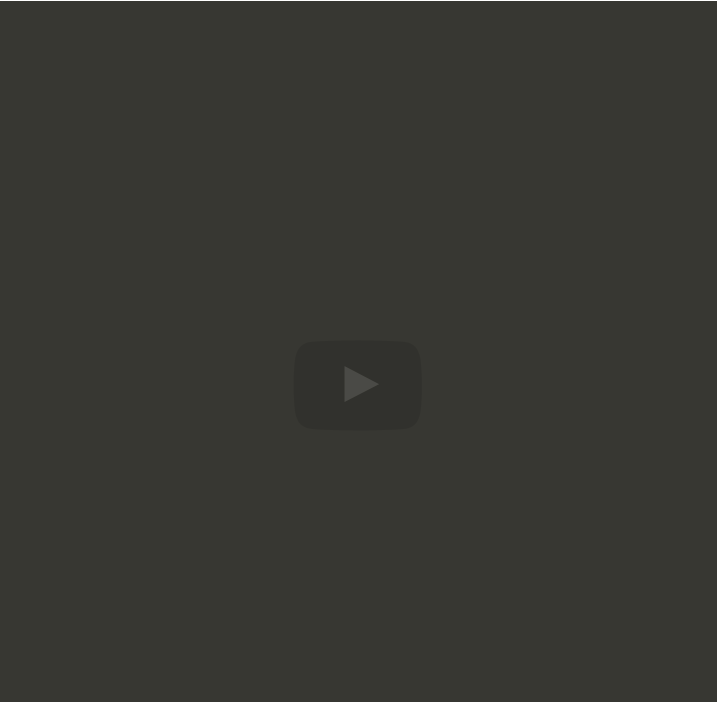
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remember that the singular values are ordered from largest to smallest, then if you go ahead and set all of these to 0 because for example you believe them to be small enough that they should have be treated as 0, the resulting matrix is the rank k matrix that in the 2-norm or the Frobenius norm is closest to matrix A. So now we have a way of detecting the rank of matrix or the effective rank of a matrix. How can we

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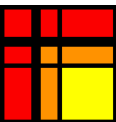
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Reading assignment

1/1 point (graded)



Read Unit 2.3.6 of the notes. [\[LINK\]](#)

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Homework 2.3.6.1

1/1 point (graded)

Given $\mathbf{A} \in \mathbb{C}^{m \times n}$, let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$ be its SVD. Show that

$$\mathbf{A}\mathbf{v}_j = \sigma_j \mathbf{u}_j \text{ for } 0 \leq j < \min(m, n),$$

where $\mathbf{u}_j$ and $\mathbf{v}_j$ equal the columns of $\mathbf{U}$ and $\mathbf{V}$ indexed by j , and σ_j equals the diagonal element of $\mathbf{\Sigma}$ indexed with m .

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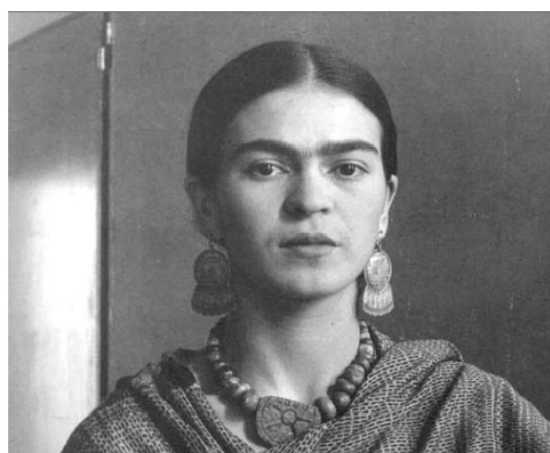
Homework 2.3.6.2

1/1 point (graded)

In `Assignments/Week02/matlab/` execute

```
> IMG = imread( 'Frida.jpg' );  
> A = double( IMG( :, :, 1 ) );  
> imshow( uint8( A ) )  
> size( A )
```

to generate the picture of Mexican artist Frida Kahlo



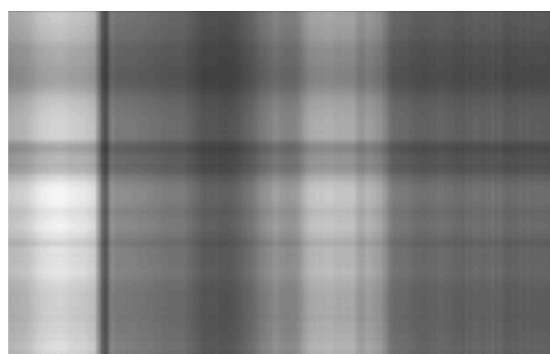
Although the picture is black and white, it was read as if it is a color image, which means a $m \times n \times 3$ array of pixel information is stored. Setting $\mathbf{A} = \text{IMG}(:, :, 1)$ extracts a single matrix of pixel information. (If you start with a color picture, you will want to approximate $\text{IMG}(:, :, 1)$, $\text{IMG}(:, :, 2)$, and $\text{IMG}(:, :, 3)$ separately.)

Next, compute the SVD of matrix $\mathbf{A}$

```
> [ U, Sigma, V ] = svd( A );
```

and approximate the picture with a rank- k update, starting with $k = 1$:

```
> k = 1  
> B = uint8( U( :, 1:k ) * Sigma( 1:k, 1:k ) * V( :, 1:k )' );  
> imshow( B );
```





Repeat this with increasing k .

```
> r = min( size( A ) )
> for k=1:r
>     imshow( uint8( U( :, 1:k ) * Sigma( 1:k,1:k ) * V( :, 1:k )' ) );
>     input( strcat( num2str( k ), "    press return" ) );
> end
```

To determine a reasonable value for k , it helps to graph the singular values:

```
> figure
> r = min( size( A ) );
> plot( [ 1:r ], diag( Sigma ), 'x' );
```

Since the singular values span a broad range, we may want to plot them with a log-log plot

```
> loglog( [ 1:r ], diag( Sigma ), 'x' );
```

For this particular matrix (picture), there is no dramatic drop in the singular values that makes it obvious what k is a natural choice.

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